

TACHS Exam Simulator

Full-Length Practice Exam 08

Complete printable test with 200 multiple-choice questions, answer key, and detailed explanations.

Section	Questions	Time
Reading	50	40 minutes
Written Expression	50	40 minutes
Mathematics	50	45 minutes
Ability	50	35 minutes

General Directions

Work through each section within the recommended time. Choose the best answer for every question. Mark one answer (A-D) for each item. Use the answer key and explanations only after completing the test.

Recommended total testing time: 160 minutes. Calculators should be used only when permitted by the rules followed for your TACHS preparation.

Section 1: Reading

50 questions | Recommended time: 40 minutes

Literary Passage: The Batch Without a Bell

Mina was helping at her uncle's bakery when the small oven timer stopped ringing. A tray of sesame rolls was already baking, and the wall clock had no second hand. Mina offered to open the oven every few minutes, but her uncle shook his head. Each opening would release heat.

He asked her what had changed since the rolls went in. Mina noticed that the dough had expanded and the pale surfaces had begun to color. A warm, nutty smell reached the counter. Her uncle tapped the oven window and pointed to the edges touching the pan, which were turning a deeper brown.

Mina wrote down the clock time when those signs appeared. For the next tray, she watched the rolls rather than waiting for a bell. They browned sooner because the oven had remained hot between batches. She recorded that difference too.

At closing time, her uncle replaced the timer's battery. Mina was relieved to hear its sharp ring again, but she no longer treated the sound as the only evidence that baking was complete. The broken timer had not taught her to ignore measurement. It had taught her to connect measured time with changes she could see and smell.

1. Why does Mina's uncle discourage opening the oven repeatedly?

- A) The wall clock would stop.
- B) The oven door is locked.
- C) The rolls need to be turned by hand.
- D) Opening it would release heat.

2. Why does the second tray brown sooner?

- A) It contains no sesame seeds.
- B) Mina opens the door more often.
- C) The oven is already hot from the first batch.
- D) The replacement timer runs too quickly.

3. Which statement best expresses a central idea of the passage?

- A) Timers are unnecessary in professional kitchens.
- B) Smell is always more reliable than time.
- C) A recipe should never be adjusted between batches.
- D) Combining measurements with direct observation leads to better judgment.

4. As used in the final paragraph, "treated" most nearly means

- A) covered with a substance.
- B) provided medical care for.
- C) gave as a gift.
- D) regarded or considered.

Informational Passage: Leaves That Send Water Away

After rain, water beads on some leaves and rolls away almost immediately. This behavior depends partly on the chemistry of the leaf surface and partly on its microscopic texture.

A leaf may be coated with waxy substances that water does not spread across easily. Under magnification, the surface may also contain tiny bumps. A droplet rests on the tops of these bumps with small pockets of air beneath it, reducing contact with the leaf. Because the droplet touches less surface, it remains rounded and rolls away more readily.

Moving droplets can collect dust and small particles as they travel. This “self-cleaning” effect may help keep surfaces clear for light or reduce the time moisture remains available for some microbes. It does not mean the leaf can remove every contaminant or prevent every disease.

Engineers study these surfaces when designing water-repellent paint, fabric, and glass. Simply adding roughness is not enough; the texture must work with suitable surface chemistry. If bumps are too large, poorly arranged, or easily damaged, water may settle between them. The natural leaf therefore offers not a single recipe but a combination of structure and material properties.

5. What may become trapped beneath a droplet on a textured leaf?

- A) Plant roots
- B) Small pockets of air
- C) Layers of soil
- D) Sunlight reflected from glass

6. Why does the author state that self-cleaning cannot remove every contaminant?

- A) To argue that leaf texture has no function
- B) To show that all microbes require dry surfaces
- C) To prove that droplets never collect dust
- D) To clarify the limits of the effect

7. How is the passage mainly organized?

- A) A surface effect is described, explained, and connected to possible applications and limits.
- B) Two scientists debate whether water exists.
- C) A leaf is followed through four seasons.
- D) Several fabrics are ranked by price.

8. Which artificial surface would most likely repel water effectively?

- A) A durable microscopic texture combined with water-repelling chemistry
- B) A rough surface made from a material that attracts water
- C) A surface with large damaged pits that hold droplets
- D) A smooth absorbent cloth with no coating

Humanities Passage: What a Doorway Can Tell Us

A doorway is designed for passage, yet its size, material, and location can reveal how a building was intended to be used. A wide entrance facing a public square invites a different kind of movement from a narrow side door opening into an alley.

Wear marks offer additional evidence. A stone threshold polished at its center may show where many feet crossed. Scratches near the bottom of a door can suggest repeated contact with carts or equipment. Such marks do not explain themselves, but they help researchers form questions about traffic and daily routines.

Doorways also change. An entrance may be widened, sealed, or moved when a building gains a new purpose. A former warehouse converted into apartments might keep a loading door as a large window. The altered opening preserves part of the earlier structure while serving a new function.

Interpreting architecture requires caution. A worn threshold could result from centuries of ordinary use or a shorter period of unusually heavy traffic. Researchers compare the doorway with plans, photographs, nearby features, and records when available. The goal is not to make a doorway “speak” with certainty, but to treat it as one source of physical evidence.

9. What might scratches near the bottom of a door suggest?

- A) That the doorway was never used
- B) The exact date the building opened
- C) Repeated contact with carts or equipment
- D) A change in the color of nearby stone

10. As used in the fourth paragraph, “caution” most nearly means

- A) speed.
- B) a warning sign beside a door.
- C) fear of entering a building.
- D) care to avoid unsupported conclusions.

11. Which statement best summarizes the passage?

- A) Doorway design, wear, and alteration can provide evidence when interpreted with other sources.
- B) Modern buildings contain no evidence of earlier use.
- C) Every doorway has one certain historical meaning.
- D) Wide entrances are always older than narrow entrances.

12. Why might a former loading door remain visible after a warehouse becomes apartments?

- A) The original doorway proves how many people lived there.
- B) Building owners are never permitted to change any opening.
- C) It can be adapted for a new use while preserving evidence of the old structure.
- D) A loading door always provides the best apartment entrance.

13. The author’s tone is best described as

- A) curious but careful about the limits of evidence.
- B) humorous and playful.
- C) critical of architectural research.
- D) certain and absolute.

Literary Passage: The Chair at the Edge

For the school portrait, the photography club arranged two neat rows of chairs in the gym. Mara checked the camera angle, then noticed that the last chair barely fit inside the frame. "Move it inward," suggested Theo, who was directing students to their places.

Mara started to drag the chair, but its metal leg caught on a strip of tape marking the boundary of the volleyball court. The scrape made everyone turn. She stopped, embarrassed, and studied the camera screen again. If she shifted the chair inward, the front row would look crowded. If she left it alone, its occupant might appear separated from the group.

Mr. Imani, the custodian, was rolling away a cart of folded mats. Without giving advice, he paused beside the chair and looked from it to the painted court line. Mara remembered how he had once measured a crooked display by comparing the empty spaces on both sides rather than staring at the display itself.

She moved the entire back row six inches toward the edge and adjusted the camera slightly. Now the spaces between the students were even, and the final chair no longer looked isolated. When Theo asked what had changed, Mara pointed to the gaps. "I was trying to fix one chair," she said. "The picture needed a different balance."

14. What first prevents Mara from moving the last chair smoothly?

- A) Theo asks her to leave it where it is.
- B) A student is already sitting in it.
- C) The photography lights block the way.
- D) Its leg catches on a strip of tape.

15. Why does Mr. Imani's pause help Mara even though he says nothing?

- A) It reminds her of a way to judge balance by examining surrounding space.
- B) It shows that he expects her to remove the court tape.
- C) It signals that the gym must be cleared immediately.
- D) It confirms that the camera should remain in its original position.

16. Which statement best expresses the central idea of the passage?

- A) A solution may emerge when a person considers the whole arrangement instead of one part.
- B) Quiet observers usually understand a problem better than participants do.
- C) Group photographs require expensive equipment to look professional.
- D) School activities work best when every participant follows the first plan.

17. The author's description of Mara's embarrassment mainly serves to

- A) criticize the students for turning toward a sudden noise.
- B) show the pressure Mara feels before she pauses and rethinks the problem.
- C) explain why Theo becomes responsible for arranging the rows.
- D) suggest that Mara no longer wants to take the photograph.

Scientific Passage: Measuring a Stream with Orange Peel

Scientists who study streams sometimes need a quick estimate of how fast surface water is moving. One simple field method uses a harmless floating object, such as a piece of orange peel. Researchers measure a straight section of the stream, release the object upstream, and record how long it takes to reach the end of the measured distance. Dividing distance by time gives the object's surface speed.

That result is not exactly the stream's average speed. Water near the banks and bottom is slowed by friction against soil, rocks, and plants. The floating object usually travels in faster water near the surface, so researchers multiply its speed by a correction factor based on the stream's conditions. They may repeat the trial several times and average the results because wind, small eddies, or an unlucky collision with a twig can affect one trip.

The method is especially useful when sophisticated instruments are unavailable or when a team needs a preliminary measurement before choosing where to collect more detailed data. It does not replace electronic flow meters for every investigation. Instead, it turns an ordinary object into a practical tool, provided that researchers recognize what the measurement represents and account for its limits.

18. Why do researchers apply a correction factor to the floating object's speed?

- A) The measured distance changes as the object moves downstream.
- B) Electronic flow meters report speed in different units.
- C) Orange peel absorbs water and becomes heavier during a trial.
- D) Surface water generally moves faster than water slowed near the banks and bottom.

19. As used in the second paragraph, "preliminary" most nearly means

- A) legally required.
- B) done as an early step.
- C) secret and protected.
- D) unusually precise.

20. How is the passage mainly organized?

- A) It explains a method, identifies sources of error, and describes appropriate uses.
- B) It compares two streams and ranks the accuracy of their measurements.
- C) It presents a historical discovery and traces its influence over time.
- D) It states a problem and argues that only electronic equipment can solve it.

21. Which conclusion is best supported by the passage?

- A) A floating object produces reliable results only in perfectly straight streams.
- B) A simple measurement can be useful when its limitations are understood.
- C) Repeating trials eliminates the need to consider friction.
- D) Wind affects water near the bottom more than water at the surface.

Humanities Passage: Margins for the Binder

Before inexpensive printing, many books were assembled from large sheets that were folded into smaller groups of pages. A craftsperson sewed these groups together and attached them to a cover. The printed text could not extend equally close to every edge. Space had to remain near the fold so words would not disappear into the binding, while other margins protected the text from fingers, trimming tools, and ordinary wear.

These blank areas also influenced how a reader experienced the page. A narrow band around dense writing could make the text seem cramped. Wider space could separate the writing from the surrounding world and give a reader's eye a place to rest. In some books, owners later filled margins with notes, corrections, pointing hands, or small drawings. The supposedly empty space became a record of attention.

Modern pages are often designed on screens, yet designers still adjust margins with similar concerns in mind. A paperback needs room near its glued spine; a digital page needs space so words do not press against the edge of a device. Although materials and tools change, margins continue to perform two roles: they protect readability through practical spacing, and they shape the visual character of a page.

22. According to the passage, why was space needed near the fold of an early book?

- A) To provide an area for illustrations only
- B) To indicate where owners should sign their names
- C) To keep printed words from disappearing into the binding
- D) To make every margin exactly the same width

23. In the second paragraph, “dense” most nearly means

- A) difficult to translate.
- B) closely packed.
- C) written in faded ink.
- D) surrounded by decoration.

24. Which statement best summarizes the passage?

- A) Digital devices have removed the need for careful page design.
- B) Handwritten notes are the main reason early books survived.
- C) Margins have long supported both the practical use and the visual design of pages.
- D) Bookbinders preferred wide margins because paper was inexpensive.

25. Why does the author discuss both printed books and digital pages?

- A) To demonstrate that the purposes of margins continue across changing formats
- B) To argue that readers should write notes on every kind of page
- C) To show that binding methods have remained completely unchanged
- D) To prove that digital text is harder to read than printed text

Literary Passage: The Unlabeled Switch

Nina had volunteered to operate the lights for the student puppet show. During rehearsal, every switch on the control board wore a paper label except one. Mr. Calder had warned her that the unlabeled switch controlled equipment used by the auditorium staff, so she should leave it alone.

Ten minutes before the audience entered, a blue spotlight stopped responding. Nina checked its cable and replaced its lamp, but the stage remained dark on the left side. Her partner, Ellis, pointed toward the unlabeled switch. "Maybe that wakes up the whole row," he said.

Nina rested her hand beside the switch without touching it. Guessing might restore the light, but it might also disturb equipment she did not understand. She opened the control-board drawer and found a laminated circuit map beneath several old programs. The map showed that the blue spotlight and two others shared a small reset button on the back panel. The mysterious switch belonged to the ventilation system.

Nina pressed the reset button, and the blue circle returned to the stage curtain. After the performance, she made a clear label for the reset and placed the circuit map in a sleeve beside the board. Ellis joked that she had solved the problem by refusing the most tempting solution. Nina smiled. "I solved it by finding out what the switches actually did."

26. What does the circuit map reveal about the unlabeled switch?

- A) It resets all three spotlights.
- B) It controls the ventilation system.
- C) It opens the stage curtain.
- D) It supplies power to the control board.

27. Why does Nina place the circuit map beside the board after the performance?

- A) She wants to hide the old programs from the auditorium staff.
- B) She expects the blue spotlight to fail during every show.
- C) She plans to redesign the lighting system herself.
- D) She hopes future operators can verify controls instead of guessing.

28. Which statement best expresses the central idea of the passage?

- A) Technical problems are usually caused by missing labels.
- B) Partners should always agree before operating equipment.
- C) Careful investigation is safer and more effective than an appealing guess.
- D) Old documents are more useful than modern equipment.

29. In the sentence "Guessing might restore the light," the word "restore" most nearly means

- A) decorate.
- B) locate.
- C) return to working condition.
- D) replace permanently.

Scientific Passage: The Funnel Builder

In loose, dry soil, an antlion larva builds a small funnel-shaped pit. It begins by moving backward in a circle while tossing grains outward with its head. As the circle narrows, a steep depression forms. The larva then buries most of its body at the bottom, leaving only its jaws near the surface.

An ant or another small insect that steps onto the sloping wall may cause grains to slide. The insect tries to climb, but the loose slope gives way beneath its feet. If the insect nears the rim, the antlion may fling additional sand upward. Falling grains strike the insect and can start another tiny slide toward the pit's center.

The pit works only under certain conditions. Damp grains cling together and do not flow readily, while grains that are too large are difficult for the larva to toss. Researchers have tested pits made with different grain sizes and found that antlions often build steeper, more effective traps in material that shifts easily.

The antlion's behavior does not make the sand unusual. Instead, the larva takes advantage of ordinary properties of loose grains. Its construction method creates a slope close to the point at which the surface begins to collapse, turning a modest movement by an insect into a larger slide.

30. How does an antlion larva begin constructing its pit?

- A) It carries damp soil into a circular mound.
- B) It moves backward in a circle and tosses grains outward.
- C) It digs a straight tunnel beneath a large stone.
- D) It presses large grains into a firm wall.

31. Why are damp grains less suitable for an antlion's pit?

- A) They are too light for the larva to control.
- B) They hold together instead of sliding easily.
- C) They hide the movement of approaching insects.
- D) They make the bottom of the pit warmer.

32. The author describes researchers' tests mainly to

- A) show that large grains always create the deepest pits.
- B) argue that antlions should be studied only in laboratories.
- C) provide evidence that grain properties affect how well the trap functions.
- D) explain how researchers protect insects from antlions.

33. How does the final paragraph contribute to the passage?

- A) It introduces a second animal that uses the same hunting method.
- B) It lists the steps for moving an antlion to a new habitat.
- C) It challenges the earlier description of how the pit is built.
- D) It explains the broader principle that makes the constructed slope effective.

Humanities Passage: Messages in Knotted Cords

Long before electronic spreadsheets, administrators in the Inca Empire used bundles of knotted cords called khipus to organize information. A main cord supported hanging cords of different colors and lengths. Knots tied at particular positions could represent quantities in a place-value system. The location and type of a knot helped a trained user distinguish ones, tens, hundreds, and larger values.

Khipus were portable and did not require flat writing surfaces. Specialists could carry them while traveling between communities, then use the recorded quantities to help track goods, labor, or population counts. Color, cord grouping, and the direction in which fibers were twisted may also have conveyed categories or relationships, though scholars continue to debate the meaning of many surviving examples.

Studying a khipu requires more than counting knots. Researchers compare cord materials, spacing, colors, and construction patterns. They also examine archaeological context and early written descriptions made after Europeans arrived in the Andes. No single clue explains every feature.

Khipus demonstrate that information can be structured without ink or paper. Their cords made numerical relationships tangible: values could be handled, grouped, and inspected. At the same time, unanswered questions about their nonnumerical features remind researchers to separate what the evidence strongly supports from what remains an informed possibility.

34. According to the passage, what helped indicate place value on a khipu?

- A) The weight of the main cord
- B) The number of people carrying it
- C) The age of the fiber
- D) The position and type of a knot

35. As used in the fourth paragraph, “tangible” most nearly means

- A) able to be touched or physically handled.
- B) difficult to preserve.
- C) written in a secret language.
- D) valuable enough to trade.

36. Which statement best summarizes the passage?

- A) Khipus were mainly decorative objects exchanged between neighboring communities.
- B) Khipus organized information through physical features, although some features remain uncertain.
- C) Researchers have fully translated every surviving khipu by counting its knots.
- D) Written descriptions are the only useful evidence about khipus.

37. Why does the passage state that “No single clue explains every feature”?

- A) To prove that cord color had no meaning
- B) To suggest that most surviving cords are modern copies
- C) To show that researchers must combine several kinds of evidence
- D) To explain why quantities cannot be recorded with knots

38. Which description best characterizes the author's tone?

- A) Amused and informal
- B) Respectful of the evidence while cautious about uncertain interpretations
- C) Critical of researchers who study material objects
- D) Doubtful that khipus had any practical use

Literary Passage: The Handle That Stayed Rough

Jonah had shaped a wooden handle for the long drawer beneath the art-room sink. He sanded it until the curved top felt as smooth as a river stone. Along the underside, however, a shallow patch of tool marks remained. He reached for finer sandpaper, determined to erase them before Ms. Vale inspected his work.

At that moment, Tessa entered carrying a stack of wet paint trays. She tried the new handle with slippery fingers. Her hand slid across the polished curve, then caught securely on the rough patch below. "That part helps," she said.

Jonah frowned. The marks looked unfinished to him. He tested the handle with a damp cloth wrapped around his fingers and noticed the same difference. Instead of sanding the patch away, he evened its edges so it felt intentional. Then he cut three narrow grooves beside it, repeating the texture across the underside.

When Ms. Vale arrived, Jonah turned the handle over before she could comment on its appearance. He explained the test and showed her the revised grip. She pulled the drawer open with a paint-smeared glove. "You didn't hide the problem," she said. "You asked whether it was a problem." Jonah looked at the smooth top and patterned underside. The handle was not the object he had first imagined, but it was better suited to the hands that would use it.

39. What causes Jonah to reconsider sanding away the tool marks?

- A) Ms. Vale asks him to add a decoration.
- B) The drawer becomes stuck beneath the sink.
- C) Tessa's slippery hand catches on the rough patch.
- D) He runs out of fine sandpaper.

40. Why does Jonah add three grooves beside the rough patch?

- A) To mark the drawer as property of the art room
- B) To make the useful texture appear deliberate and consistent
- C) To remove a crack running through the handle
- D) To copy the shape of Tessa's paint trays

41. Ms. Vale's final comment emphasizes that Jonah has learned to

- A) examine an apparent flaw before deciding how to correct it.
- B) value appearance more than practical use.
- C) finish a project without asking anyone for advice.
- D) follow his original plan exactly.

42. As used in the third paragraph, "intentional" most nearly means

- A) temporary.
- B) done by design.
- C) copied from another object.
- D) difficult to notice.

Scientific Passage: Reading Vibrations in a Web

A spiderweb is more than a trap. Its threads also carry vibrations that can tell a spider what is happening at a distance. An insect striking the web produces a pattern of movement that travels through several strands. Wind, falling leaves, and the spider's own steps produce vibrations too, but their patterns differ in strength, rhythm, and direction.

Many web-building spiders rest with their legs touching selected threads. Sensory organs near their leg joints detect tiny changes in tension. By comparing signals arriving through different legs, a spider can often determine which region of the web is moving. It may then approach a trapped insect, ignore harmless motion, or retreat from a disturbance that suggests danger.

The web's condition affects these signals. A strand that has stretched or broken may transmit movement differently. Some spiders regularly tighten, replace, or consume old silk and build new sections. These actions help maintain both the web's physical shape and its usefulness as a source of information.

Researchers study web vibrations with sensitive instruments that record thread movement. Their work shows that the meaning of a signal depends not simply on whether the web shakes, but on how it shakes. The spider's success relies on detecting differences within a busy mixture of motions.

43. Where are the sensory organs that detect changes in thread tension located?

- A) Near the spider's leg joints
- B) Along the spider's abdomen
- C) At the center of the web
- D) Inside newly produced silk

44. Which statement best summarizes the passage?

- A) Spiders use varied vibration patterns in maintained webs to gather information.
- B) Spiders respond to every web movement in the same way.
- C) Wind is the main cause of damage to spiderwebs.
- D) Researchers build artificial webs to capture insects.

45. In the third paragraph, "transmit" most nearly means

- A) measure precisely.
- B) block completely.
- C) create from nothing.
- D) carry from one place to another.

46. Why does the author mention wind, falling leaves, and the spider's steps?

- A) To show that a spider must distinguish useful signals from other movement
- B) To prove that insects rarely create noticeable vibrations
- C) To explain why spiders abandon every damaged web
- D) To list the materials spiders use to produce silk

Humanities Passage: Writing Beneath Writing

In periods when prepared parchment was costly, a sheet might be used more than once. A scribe could scrape or wash away an earlier text and write a new one on the same surface. Such a reused manuscript page is called a palimpsest. To an ordinary reader, the newer writing may dominate the page, while faint traces of the older writing remain beneath it.

Those traces can preserve works that would otherwise be unknown. Scholars once tried to read them by changing the angle of light or applying chemicals, but some chemicals stained or weakened the parchment. Today, imaging systems can photograph a page under several wavelengths of light. Because inks and parchment respond differently to different wavelengths, computer processing may make faded letters easier to distinguish.

Recovering an earlier text is not as simple as revealing a hidden layer all at once. Marks from the two writings can cross, letters may be missing, and damage may resemble ink. Specialists compare repeated letter forms and possible word patterns, then record how certain or uncertain each reading is.

A palimpsest therefore holds more than one moment of use. Its newest text shows what a later owner considered worth recording, while the older traces reveal what the material carried before. The page is both a document and evidence of changing needs.

47. Why did scribes sometimes remove an earlier text from parchment?

- A) New writing adhered only to scraped letters.
- B) Older ink prevented pages from being folded.
- C) Every manuscript required two layers of writing.
- D) Parchment was costly and could be reused.

48. Why do specialists record how certain they are about recovered words?

- A) Modern imaging produces only black-and-white photographs.
- B) The newer text is always less important than the older one.
- C) Overlapping marks, missing letters, and damage can make readings uncertain.
- D) Scholars are not permitted to compare repeated letter forms.

49. What is the central idea of the passage?

- A) Reused parchment can preserve layered evidence that modern methods help scholars interpret.
- B) Chemical treatment is the safest way to study every manuscript.
- C) The newest writing on a manuscript is usually impossible to read.
- D) Scribes reused parchment mainly to make books thinner.

50. How is the passage mainly organized?

- A) It ranks several manuscripts from least to most valuable.
- B) It presents two opposing arguments about using parchment.
- C) It defines a type of page, explains how it is studied, and considers what it reveals.
- D) It follows one scribe through the complete production of a book.

Section 2: Written Expression

50 questions | Recommended time: 40 minutes

Revision Passage: Monitoring a Compost Pile

[1] Our school garden started a compost pile for fruit scraps and dry leaves. [2] To understand how the pile changes, students record its internal temperature twice each week. [3] We insert the same thermometer to the same depth and wait one minute before reading it. [4] The temperature rose during the first two weeks as microbes broke down the material. [5] The pile is located behind the storage shed. [6] When the center cooled, we mixed the outer material inward and added a small amount of water. [7] Keeping a consistent record helps us connect each change in temperature with the actions taken. [8] The log will also help next year's class compare its composting process with ours.

51. Which sentence should be removed because it does not develop the monitoring process?

- A) Sentence 5
- B) Sentence 3
- C) Sentence 2
- D) Sentence 7

52. Which revision most effectively combines sentences 2 and 3?

- A) Using a thermometer and because the pile changes, twice weekly at the same depth.
- B) Students record temperature twice, the thermometer same depth, waiting one minute each week.
- C) Twice each week, students insert the same thermometer to the same depth, wait one minute, and record the compost's internal temperature.
- D) The same thermometer, students wait and record, it is inserted to a depth.

53. Which phrase best replaces "the actions taken" in sentence 7?

- A) the stuff that we did to it
- B) whatever was done during that time
- C) various actions of different kinds
- D) mixing the pile or adding water

54. Which sentence would best follow sentence 4?

- A) Microbial activity often decreases as available material changes.
- B) The garden fence was painted last spring.
- C) Students prefer apples to pears at lunch.
- D) The storage shed contains several rakes.

Revision Passage: A Student Art Lending Program

[1] The school gallery displays more student artwork than can fit on its walls at one time. [2] The art club proposes lending framed pieces to classrooms for six-week periods. [3] Teachers could choose from a digital catalog that lists each work's dimensions and handling instructions. [4] A condition form would be completed before and after every loan. [5] Each frame have a label with its identification number. [6] Borrowers would carry larger pieces with two people and avoid placing them in direct sunlight. [7] The program would allow more students to see the collection while protecting the artwork through clear procedures. [8] After the first semester, the club could review condition forms and revise the rules if needed.

55. Which revision corrects the error in sentence 5?

- A) Each frame have a label with its identification number.
- B) Each frame has a label with its identification number.
- C) Each frames has a label with its identification number.
- D) Each frame having a label with its identification number.

56. Which revision most effectively combines sentences 4 and 5?

- A) Condition forms before and after, each frame having a number for loans.
- B) Before and after each loan, borrowers would complete a condition form for the framed work identified by its numbered label.
- C) The numbered label and a condition form, borrowers before and after.
- D) Because every loan has a frame, forms being completed and labels.

57. Which revision of sentence 6 is most clear and concise?

- A) Two people would carry large pieces, and borrowers would keep all artwork out of direct sunlight.
- B) For large art, two people should both carry it, and nobody should put any of it where sun is direct.
- C) Borrowers, with pieces large and sunlight direct, would use two people and avoidance.
- D) Larger pieces would be carried by two people together, and direct sunlight would be avoided by borrowers.

58. Which sentence would provide the best opening before sentence 1?

- A) Some classrooms have windows facing east.
- B) The gallery walls were painted white during summer.
- C) Student artwork can continue serving an audience even when it is not in the main gallery.
- D) Frames are available in many colors and materials.

59. Which sentence is written correctly?

- A) The number of applicants has increased.
- B) A number of applicants has arrived early.
- C) The applicants number have increased.
- D) The number of applicants have increased.

60. Which sentence is punctuated correctly?

- A) After the meeting we reviewed, the notes.
- B) After the meeting, we reviewed the notes.
- C) After, the meeting we reviewed the notes.
- D) After the meeting we, reviewed the notes.

61. Which sentence makes the comparison most clearly and correctly?

- A) This route is safer than the old one.
- B) This route is safest than the old one.
- C) This route is more safer than the old one.
- D) This route is more safest than the old one.

62. Which numbered line contains an error in standard written US English?

Line 1: The wind pushed the empty boat from shore.

Line 2: A rope kept it from drifting into the channel.

Line 3: We pulled the boat back with no difficulty.

Line 4: Its wooden seats was still dry.

- A) Line 4
- B) Line 2
- C) Line 3
- D) Line 1

63. Which numbered line contains an error in standard written US English?

Line 1: Mara offered the extra ticket to Luis and I.

Line 2: We arrived at the theater before the doors opened.

Line 3: An usher directed us to the balcony.

Line 4: The performance began exactly on time.

- A) Line 2
- B) Line 1
- C) Line 3
- D) Line 4

Revision Passage: Calibrating the Hallway Scale

[1] Our science club discovered that the hallway scale gave a different reading each time we weighed the same sealed container. [2] Before deciding that the scale was broken, we placed it on a hard, level tile instead of the soft entrance mat. [3] We also checked that the pointer returned to zero after each trial. [4] Next, the container was weighed five times, and every result was recorded. [5] The readings became consistent once the scale stood on the tile. [6] This investigation taught us that a measuring tool's surroundings can influence its results. [7] The custodian unlocks the building at 6:45 each morning. [8] We posted a small sign reminding users to keep the scale on a firm surface.

64. Which sentence should be removed because it does not support the passage's central idea?

- A) Sentence 2
- B) Sentence 7
- C) Sentence 5
- D) Sentence 8

65. Which revision of sentence 4 is clearest and most direct?

- A) Next, there were five weighings of the container that were recorded by us.
- B) Next, the container, after that, had five results and recordings.
- C) Next, five times was how the container received its recorded weighing.
- D) Next, we weighed the container five times and recorded every result.

66. Which sentence best combines sentences 5 and 6 while preserving their meaning?

- A) The readings became consistent, this taught us the surroundings matter.
- B) Becoming consistent, the surroundings influenced the measuring tool and taught us.
- C) The readings became consistent on the tile, learning about a measuring tool's surroundings.
- D) Because the readings became consistent on the tile, we learned that a measuring tool's surroundings can influence its results.

**67. Where should the following sentence be placed to best clarify the club's first step?
First, we marked the container so that everyone used the same object.**

- A) Before sentence 1
- B) After sentence 1
- C) After sentence 6
- D) After sentence 3

Revision Passage: A Guide to Quiet Study Signals

[1] Students in our after-school study room needed a way to request help without interrupting others. [2] We created three small desk cards: green meant "working independently," yellow meant "question when available," and red meant "help needed now." [3] During a one-week trial, tutors noted how often each card was used. [4] The red card was used less often after students became familiar with the system. [5] For example, several students used yellow when they could continue another problem while waiting. [6] The cards were laminated, this made them durable enough for daily use. [7] At the end of the trial, students suggested adding brief explanations to the back of each card. [8] Their suggestion made the system easier for new participants to learn.

68. Which revision of sentence 2 uses punctuation correctly?

- A) We created three small desk cards green meant "working independently" yellow meant "question when available" and red meant "help needed now."
- B) We created three small desk cards; green meant "working independently," yellow meant "question when available," and red meant "help needed now."
- C) We created three small desk cards, green meant "working independently"; yellow meant "question when available"; and red meant "help needed now."
- D) We created three small desk cards: green meant "working independently," yellow meant "question when available," and red meant "help needed now."

69. Which revision of sentence 4 most precisely explains the change?

- A) Students became familiar, and the red card was a card used differently.
- B) The red card changed in a way that was less after students knew about things.
- C) As students learned when to use each signal, they reserved the red card for problems that stopped their progress.
- D) After a while, it was not used as much for various reasons.

70. Which revision best corrects sentence 6?

- A) The cards being laminated and durable enough for daily use.
- B) Laminated, daily use made the cards durable enough.
- C) The cards were laminated, being durable enough for daily use.
- D) The cards were laminated, which made them durable enough for daily use.

71. Which sentence would best conclude the passage?

- A) Tutors often carry pens and notebooks to the study room.
- B) Red, yellow, and green are also common colors on traffic signals.
- C) A simple visual system can support both concentration and timely assistance.
- D) Some students prefer to complete homework before dinner.

72. Which sentence uses the comparative form correctly?

- A) Of the two routes, the eastern path is shortest.
- B) This notebook is more thinner than the blue one.
- C) Ava's estimate was the most closer of the two.
- D) The second sample is more stable than the first sample.

73. Which sentence uses the correct pronoun case?

- A) The coach gave Mia and I the updated schedule.
- B) The librarian asked Noah and me to close the windows.
- C) The final decision belongs to she and Marcus.
- D) Her and Luis organized the supply shelf.

74. Which line contains an error in standard written US English?

Line 1: The workshop includes three stations.

Line 2: Each station offers a different task.

Line 3: Participants may choose whichever activity interests them.

Line 4: Neither of the instructors are available before noon.

- A) Line 3
- B) Line 2
- C) Line 4
- D) Line 1

75. Which line contains an error in standard written US English?

Line 1: By the time the guests arrived, we had arranged the chairs.

Line 2: The speaker begins her presentation after everyone was seated.

Line 3: She then answered questions from the audience.

Line 4: The meeting ended at the scheduled time.

- A) Line 1
- B) Line 4
- C) Line 3
- D) Line 2

76. Which line contains an error in standard written US English?

Line 1: The ceramic bowl was packed carefully.

Line 2: A layer of paper protected its rim.

Line 3: The box was labeled fragile before it was carried downstairs.

Line 4: The courier handled the package careful.

- A) Line 1
- B) Line 2
- C) Line 4
- D) Line 3

Revision Passage: Testing Window Shades

[1] Our classroom became uncomfortably warm during sunny afternoons. [2] The environmental club tested three removable window shades made from different materials. [3] Before each trial, students recorded the room temperature and closed one shade over the same window. [4] They measured the temperature again after thirty minutes. [5] The silver fabric produced the smallest temperature increase. [6] It was shiny and measured by us carefully. [7] The club also noted that the fabric could be rolled and stored without bending. [8] Based on both results, we recommended a two-week trial in another classroom.

77. Which sentence would best follow sentence 2 to explain how the test was kept fair?

- A) The classroom contains twenty-four student desks.
- B) Several club members enjoy bright summer weather.
- C) Each shade covered the same area of glass during its trial.
- D) Window shades are sold in many colors and patterns.

78. Which revision of sentence 6 is most effective?

- A) We also observed that the silver surface reflected much of the incoming light.
- B) The silver fabric was shiny, measuring it carefully.
- C) Shiny and carefully, we measured the silver fabric.
- D) Being shiny, careful measurement was done by us.

79. Which revision of sentence 8 most clearly identifies the two results?

- A) Based on things we found, we recommended trying it someplace else.
- B) Because the fabric limited warming and remained easy to store, we recommended a two-week trial in another classroom.
- C) Based on both of those results that happened, another classroom was recommended by us.
- D) We recommended a trial, and the two results were both good ones.

80. Which sentence should be placed immediately before sentence 5 to improve the presentation of results?

- A) Some students brought lunch from home that day.
- B) The club's first meeting was held in September.
- C) Warm classrooms can make concentration difficult.
- D) We compared the changes recorded for all three materials.

Revision Passage: Reusable Name Badges

[1] The welcome committee printed new paper name badges for every school event. [2] Most badges were discarded after only a few hours. [3] To reduce this waste, the committee purchased clear holders and printed reusable name cards. [4] Volunteers collect the holders at the exit, wiped them clean, and returned them to a labeled box. [5] A small sign near the door reminded guests to return their badges. [6] During the first event, forty-six of the fifty holders came back. [7] The committee replaced the four missing holders but continued using the original cards. [8] This practical change reduced repeated printing and simplified preparation for later events.

81. Which revision correctly fixes the verb tense in sentence 4?

- A) Volunteers collecting the holders at the exit, wipe them clean, and return them.
- B) Volunteers collected the holders at the exit, wiped them clean, and returned them to a labeled box.
- C) Volunteers collect the holders at the exit, had wiped them clean, and returned them.
- D) Volunteers will collect the holders at the exit, wiped them clean, and return them.

82. Which sentence best combines sentences 1 and 2?

- A) The welcome committee printed new paper name badges for every event, most were discarded after a few hours.
- B) Although the welcome committee printed new paper name badges for every event, most were discarded after only a few hours.
- C) The welcome committee printed new paper name badges, being discarded after every event.
- D) Printing new paper name badges for every event, after only a few hours.

83. Which revision of sentence 7 is most concise without changing its meaning?

- A) There were four missing holders that were replaced, but the cards originally continued in use by them.
- B) The committee did the replacing of the four holders that were missing while continuing the using of cards.
- C) Four holders were missing, so the committee replaced them and reused the remaining name cards.
- D) The original cards were cards they continued using after replacing holders, of which four were missing.

84. Which sentence uses parallel structure correctly?

- A) The guide asked us to label the samples, recording the weights, and to store the boxes.
- B) The guide asked us for labeling the samples, to record the weights, and storing the boxes.
- C) The guide asked us to label the samples, record the weights, and store the boxes.
- D) The guide asked us to label the samples, the weights were recorded, and store the boxes.

85. Which sentence uses a pronoun with a clear antecedent?

- A) The bicycles were moved away from the gate because they blocked the entrance.
- B) When Lena placed the pitcher beside the bowl, it was nearly full.
- C) After speaking with Maya, Sofia said she would revise the poster.
- D) The jacket was laid on the chair, but it had a torn leg.

86. Which sentence uses punctuation correctly?

- A) Before the bus arrived, we checked the list one final time.
- B) Before, the bus arrived we checked the list one final time.
- C) Before the bus arrived we checked the list, one final time.
- D) Before the bus arrived we checked, the list one final time.

87. Which line contains an error in standard written US English?

Line 1: The recipe requires fewer ingredients than we expected.

Line 2: Only a small amount of flour remained in the container.

Line 3: We needed less measuring cups than the other group.

Line 4: Each student cleaned one section of the counter.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

88. Which line contains an error in standard written US English?

Line 1: Walking across the courtyard, the banner caught Elena's attention.

Line 2: She stopped to read the announcement.

Line 3: The meeting was scheduled for Thursday afternoon.

Line 4: Elena added the event to her calendar.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

Revision Passage: Lending Field Magnifiers

[1] The nature club owned twelve hand magnifiers, but students often forgot which ones had been borrowed. [2] We assigned each magnifier a number and attached a matching tag to its case. [3] Before leaving for the courtyard, a student wrote the magnifier's number beside their name on a checkout sheet. [4] Returning students placed the magnifiers in a shallow tray. [5] A club monitor checked the lenses for dust and returned each tool to its numbered case. [6] The trays were blue. [7] During the first month, no magnifiers were lost, and small scratches were noticed before they became larger. [8] The simple record made the equipment easier to share and maintain.

89. Which sentence should be removed because it does not develop the passage's central idea?

- A) Sentence 3
- B) Sentence 5
- C) Sentence 6
- D) Sentence 8

90. If sentence 2 were revised to refer to all twelve magnifiers and their cases, which version would be correct?

- A) We assigned the magnifiers number's and attached matching tags to there cases.
- B) We assigned the magnifier's numbers and attached matching tags to it's cases.
- C) We assigned numbers to the magnifiers and attached matching tags to their cases.
- D) We assigned the magnifiers' numbers and attached matching tags to its cases.

91. Which revision of sentence 7 is most precise?

- A) During the first month, things went well with all of the tools and marks.
- B) The first month was good because nothing bad happened to any magnifier.
- C) For one month, no magnifiers were lost, and scratches were a thing noticed earlier.
- D) In the first month, the system prevented losses and helped the club identify minor damage promptly.

92. Which sentence best combines sentences 4 and 5?

- A) Returning students placed the magnifiers in a tray, a monitor checked and returned them.
- B) Students returned, the monitor checked the lenses, the cases were numbered.
- C) After returning students placed the magnifiers in a shallow tray, a club monitor checked the lenses and returned each tool to its numbered case.
- D) Placing magnifiers in a shallow tray, the lenses were checked by a monitor returning cases.

Revision Passage: A Waiting List for the Practice Room

[1] The music practice room could hold only one small group at a time. [2] Students once waited outside the door without knowing whose turn was next. [3] The student council placed a dry-erase schedule beside the room. [4] Groups entered their names in fifteen-minute blocks, they erased each entry after finishing. [5] An empty block remained available to any group that arrived without a reservation. [6] The council reviewed the schedule after two weeks. [7] Most groups began arriving near the start of their chosen blocks, and the hallway became less crowded. [8] A printed reminder asked users to leave promptly when their time ended.

93. Which revision best corrects sentence 4?

- A) Groups entered their names in fifteen-minute blocks, erasing each entry after finishing.
- B) Groups entering their names in fifteen-minute blocks and erased each entry after finishing.
- C) Groups entered their names in fifteen-minute blocks, they erased each entry after finishing.
- D) Entering names in fifteen-minute blocks, each entry erased after groups finished.

94. Which sentence would best follow sentence 6 to introduce the evidence from the review?

- A) Dry-erase markers are available in several colors.
- B) The council compared hallway observations from before and after the schedule was posted.
- C) Music can be performed by groups of many sizes.
- D) Some classrooms are located on the second floor.

95. Which revision of sentence 5 is most concise and clear?

- A) There was an empty block available in its availability for any such group.
- B) An empty block that was empty remained for a group without any reservation already made.
- C) Groups arriving without reservations had availability that was in an empty block.
- D) Any unreserved group could use an open time block.

96. Which sentence uses modifier forms correctly?

- A) The near completely model stood on the shelf.
- B) The nearly complete model stood on the shelf.
- C) The nearly completion model stood on the shelf.
- D) The complete nearly model stood on the shelf.

97. Which sentence uses the correct verb form?

- A) The committee has chosen three finalists.
- B) The committee had choosing three finalists.
- C) The committee has chose three finalists.
- D) The committee has choose three finalists.

98. Which line contains an error in standard written US English?

Line 1: The two proposals address the same concern.

Line 2: This proposal is more clearer than the earlier one.

Line 3: Its examples are also more specific.

Line 4: The committee will discuss both versions tomorrow.

- A) Line 1
- B) Line 2
- C) Line 3
- D) Line 4

99. Which line contains an error in standard written US English?

Line 1: Marco set the sealed envelope on the counter.

Line 2: He checked the address against his list.

Line 3: Then he laid the list beside the envelope.

Line 4: The envelope laid there until the courier arrived.

A) Line 1

B) Line 2

C) Line 3

D) Line 4

100. Which line contains an error in standard written US English?

Line 1: The museum displayed several early cameras.

Line 2: One of the devices were missing its leather case.

Line 3: A label described how the device had been used.

Line 4: Visitors could view the case in a nearby cabinet.

A) Line 1

B) Line 4

C) Line 3

D) Line 2

Section 3: Mathematics

50 questions | Recommended time: 45 minutes

101. What is the value of $3\frac{1}{2} \times 2\frac{2}{7}$?

- A) $8\frac{1}{2}$
- B) $7\frac{1}{2}$
- C) 8
- D) 9

102. Solve $8x + 5 = 61$.

- A) 6
- B) 7
- C) 8
- D) 9

103. The data are 4, 7, 9, 9, 11, and 14. What is the range?

- A) 10
- B) 7
- C) 5
- D) 18

104. A pentagon has interior angles of 100° , 105° , 110° , 115° , and x° . What is x ?

- A) 90°
- B) 100°
- C) 120°
- D) 110°

105. A 2.4-kilometer trail is divided into 6 equal sections. How long is each section?

- A) 0.4 km
- B) 0.04 km
- C) 1.4 km
- D) 4 km

106. Which value equals 125% of 64?

- A) 72
- B) 76
- C) 80
- D) 96

107. The sequence is 81, 27, 9, 3. Which rule produces each term after the first?

- A) Subtract 54.
- B) Multiply by 3.
- C) Subtract consecutive multiples of 9.
- D) Divide by 3.

108. A bag has 3 red, 4 blue, 2 green, and 1 yellow bead. What is the probability of blue or green?

- A) $\frac{3}{5}$
- B) $\frac{1}{2}$
- C) $\frac{2}{5}$
- D) $\frac{7}{10}$

109. A square has area 121 square units. What is its perimeter?

- A) 22 units
- B) 44 units
- C) 121 units
- D) 33 units

110. A machine fills 35 bottles in 5 minutes. At the same rate, how many bottles in 18 minutes?

- A) 126
- B) 119
- C) 112
- D) 140

111. What is the greatest common factor of 54 and 90?

- A) 6
- B) 18
- C) 9
- D) 27

112. Which inequality represents “four times a number decreased by 7 is less than 21”?

- A) $4n - 7 < 21$
- B) $4(n - 7) < 21$
- C) $7 - 4n < 21$
- D) $4n + 7 > 21$

113. A number is multiplied by 0.4 and then divided by 0.05. The result is 56. What is the original number?

- A) 0.7
- B) 4.48
- C) 7
- D) 112

114. The mean of five measurements is 18. Four of the measurements are 11, 16, 19, and 24. What is the fifth measurement?

- A) 22
- B) 18
- C) 20
- D) 25

115. Two angles form a linear pair. One angle measures 18° less than twice the other angle. What is the measure of the smaller angle?

- A) 54°
- B) 108°
- C) 72°
- D) 66°

116. A container holds 2.4 liters of water. After 650 milliliters are removed, how many milliliters remain?

- A) 1,750 mL
- B) 1,850 mL
- C) 1,650 mL
- D) 2,350 mL

117. What is the value of $\frac{3}{5} \div 1\frac{1}{2} + \frac{1}{4}$?

- A) $1\frac{3}{20}$
- B) $\frac{13}{20}$
- C) $\frac{19}{20}$
- D) $\frac{7}{10}$

118. For a function, the output equals 3 less than the square of the input. What is the output when the input is -4 ?

- A) -19
- B) 5
- C) 13
- D) 19

119. A bag contains 4 copper tokens, 3 silver tokens, and 5 wooden tokens. Two tokens are selected at random without replacement. What is the probability that both are silver?

- A) $\frac{3}{22}$
- B) $\frac{1}{16}$
- C) $\frac{1}{22}$
- D) $\frac{1}{4}$

120. A right triangle has legs of length 9 units and 12 units. A square is constructed on the triangle's hypotenuse. What is the area of the square?

- A) 144 square units
- B) 169 square units
- C) 196 square units
- D) 225 square units

121. Two groups recorded the following times in minutes.

Group P: 8, 8, 9, 10, 15

Group Q: 6, 9, 10, 10, 15

Which statement is true?

- A) The groups have the same mean and the same range.
- B) Group P has a greater mean and a smaller range.
- C) Group Q has a greater mean, but the groups have the same range.
- D) The groups have the same mean, but Group P has a smaller range.

122. A rectangular label measures 8 centimeters by 5 centimeters. A border 1 centimeter wide is printed entirely inside the label along all four edges. What area remains inside the border?

- A) 24 cm^2
- B) 18 cm^2
- C) 26 cm^2
- D) 40 cm^2

123. What is the least positive integer that is divisible by both 8 and 15?

- A) 30
- B) 120
- C) 60
- D) 240

124. The first four terms of a sequence are 4, 7, 13, 25. Each term after the first is formed by applying the same rule to the previous term. What is the next term?

- A) 37
- B) 43
- C) 49
- D) 51

125. A regular polygon has an exterior angle measuring 24° . How many sides does the polygon have?

- A) 12
- B) 14
- C) 18
- D) 15

126. What is the value of $2.75 - 1\frac{3}{8}$?

- A) $1\frac{1}{8}$
- B) $1\frac{5}{8}$
- C) $1\frac{3}{8}$
- D) $2\frac{1}{8}$

127. The perimeter of an isosceles triangle is 47 centimeters. Each equal side is 4 centimeters longer than the base. What is the length of the base?

- A) 11 cm
- B) 13 cm
- C) 15 cm
- D) 12 cm

128. A spinner has eight equal sections numbered 1 through 8. What is the probability of spinning a number that is either a factor of 8 or greater than 6?

- A) $\frac{1}{2}$
- B) $\frac{3}{4}$
- C) $\frac{7}{8}$
- D) $\frac{5}{8}$

129. The angles of a triangle are in the ratio 2:3:4. What is the measure of the largest angle?

- A) 60°
- B) 80°
- C) 70°
- D) 40°

130. A walking route is 3.6 kilometers long. Priya has completed 2,250 meters. How many meters remain?

- A) 1,450 m
- B) 1,250 m
- C) 1,350 m
- D) 5,850 m

131. A store reduces a \$64 item by 25% and then applies a \$6 coupon. What is the final price before tax?

- A) \$42
- B) \$46
- C) \$48
- D) \$52

132. Which value of x makes $5(2x - 3) = 4x + 21$ true?

- A) 4
- B) 5
- C) 7
- D) 6

133. The data values 6, 9, 11, 13, and n have a median of 11 and a range of 12. Which value could n equal?

- A) 4
- B) 15
- C) 18
- D) 17

134. A circle has a circumference of 18π centimeters. What is its area?

- A) $81\pi \text{ cm}^2$
- B) $36\pi \text{ cm}^2$
- C) $72\pi \text{ cm}^2$
- D) $18\pi \text{ cm}^2$

135. A machine fills 14 containers in 8 minutes at a constant rate. At that rate, how many containers can it fill in 28 minutes?

- A) 42
- B) 49
- C) 52
- D) 56

136. Which expression has the greatest value?

- A) $5/6 \times 3/5$
- B) 0.55
- C) $2/3 \div 4/3$
- D) $7/8 - 1/4$

137. The points (2, 5), (4, 9), and (7, 15) lie on the same line. Which equation describes the line?

- A) $y = x + 3$
- B) $y = 3x - 1$
- C) $y = 2x - 1$
- D) $y = 2x + 1$

138. What is the value of $-18 \div 3 + 4(-2)$?

- A) -14
- B) -2
- C) 2
- D) 14

139. A box contains 5 red clips, 4 blue clips, and 3 green clips. One clip is selected at random. What is the probability that it is not blue?

- A) $1/3$
- B) $5/12$
- C) $2/3$
- D) $3/4$

140. A rectangle has an area of 96 square units. Its length is 4 units greater than its width. What is its perimeter?

- A) 32 units
- B) 40 units
- C) 44 units
- D) 48 units

141. If $3x + 7 \leq 25$, which value of x does not satisfy the inequality?

- A) 4
- B) 5
- C) 6
- D) 7

142. On a map, 1 inch represents 18 miles. Two towns are $3 \frac{1}{4}$ inches apart on the map. What is their actual distance?

- A) 56.5 miles
- B) 58.5 miles
- C) 54 miles
- D) 60 miles

143. The prime factorization of a number is $2^3 \times 3 \times 7$. Which number is a factor of the original number?

- A) 56
- B) 48
- C) 54
- D) 36

144. The mean of six scores is 14. When one score is removed, the mean of the remaining five scores is 12. What score was removed?

- A) 18
- B) 20
- C) 22
- D) 24

145. A right rectangular prism measures 3 units by 4 units by 8 units. What is its total surface area?

- A) 96 square units
- B) 136 square units
- C) 112 square units
- D) 144 square units

146. A sequence begins 81, 27, 9, 3. If the same pattern continues, what is the seventh term?

- A) $1/9$
- B) $1/3$
- C) $1/27$
- D) 0

147. A cyclist travels 24 miles in 1.5 hours and then 18 miles in 1.5 hours. What is the cyclist's average speed for the entire trip?

- A) 12 miles per hour
- B) 13 miles per hour
- C) 16 miles per hour
- D) 14 miles per hour

148. A quantity is increased by 20% and becomes 78. What was the original quantity?

- A) 65
- B) 62.4
- C) 64
- D) 60

149. The values 4, 7, 7, 10, 12, and 14 are listed. Which value can be added so that the median becomes 9?

- A) 5
- B) 7
- C) 9
- D) 12

150. Each interior angle of a regular polygon measures 150° . How many sides does the polygon have?

- A) 10
- B) 11
- C) 12
- D) 15

Section 4: Ability

50 questions | Recommended time: 35 minutes

151. Complete the matrix. Across each row, the two symbols move one position clockwise around four corners.

Row 1 left: ○ top-left; □ bottom-right

Row 1 right: ○ top-right; □ bottom-left

Row 2 left: △ top-right; ◇ bottom-left

Row 2 right: ?

- A) △ top-right; ◇ bottom-left
- B) △ top-left; ◇ bottom-right
- C) △ bottom-left; ◇ top-right
- D) △ bottom-right; ◇ top-left

152. Complete the matrix. Down each column, the lower cell adds an outer square around the symbol.

Top-left: ○

Bottom-left: ○ inside □

Top-right: △

Bottom-right: ?

- A) □ inside △
- B) △ beside □
- C) ▲ inside □
- D) △ inside □

153. Complete the matrix. In each row, the right cell changes + to × and duplicates the shape.

Row 1: ○ + | ○ ○ ×

Row 2: □ + | ?

- A) □ × ×
- B) □ □ ×
- C) ■ ■ ×
- D) □ □ +

154. Complete the matrix. Across each row, the line is reflected left-to-right while the shape changes from open to filled.

Row 1: / ○ | \ ●

Row 2: \ □ | ?

- A) / □
- B) \ ■
- C) \ □
- D) / ■

155. Complete the matrix. The third cell shows the symbol that appears most often in the first two groups.

Row 1: ○ ○ □ | ○ △ | ○

Row 2: + ★ ★ | ★ × | ?

- A) ×
- B) +
- C) ★
- D) ☆

156. A square is folded from left to right. A hole is punched near the folded paper's top edge and another near its bottom edge, at different horizontal positions. What appears after unfolding?

- A) Two holes in one vertical line
- B) Eight holes around the edges
- C) Four holes in one vertical column
- D) Four holes forming two horizontal mirrored pairs

157. A rectangle is folded from top to bottom and then left to right. A hole is punched near the folded paper's lower-right area, away from edges. What appears after unfolding?

- A) Two holes on the right
- B) Four holes in one horizontal row
- C) Four holes in corresponding positions across both centerlines
- D) Eight holes near the corners

158. A square is folded along the diagonal from lower-left to upper-right. A small triangular cut is made on the folded diagonal edge, away from corners. What appears after unfolding?

- A) Two triangular cuts on outer edges
- B) One symmetric diamond-shaped opening centered on the diagonal
- C) One triangular cut on the diagonal
- D) Four cuts around the center

159. A square is folded from bottom to top. Three holes are punched in a horizontal row on the folded paper, away from edges. What appears after unfolding?

- A) Six holes in one horizontal row
- B) Three holes on the horizontal centerline
- C) Six holes forming three vertical mirrored pairs
- D) Twelve holes in four rows

160. The following groups share one rule:

1. ○ □ ●

2. △ ◇ ▲

3. ☆ ○ ★

Which choice follows the same rule?

- A) ■ □ ○
- B) ○ ● □
- C) □ △ ■
- D) ◇ △ ◆

161. The following arrangements share one rule:

1. Left +; center ○; right +

2. Left ×; center □; right ×

3. Left ★; center △; right ★

Which choice follows the same rule?

- A) Left ◇; center ☆; right ◇
- B) Left ○; center +; right ○
- C) Left +; center ■; right +
- D) Left +; center ○; right ×

162. The following figures share one rule:

1. Outer ○; inner + at top

2. Outer □; inner × at top

3. Outer △; inner ★ at top

Which choice follows the same rule?

- A) Outer ◇; inner ○ at top
- B) Outer ◇; inner ○ at bottom
- C) Outer ◇; inner ◆ at top
- D) Outer ■; inner + at top

163. Complete the 2×2 matrix.

Top-left: ○ | ▲

Top-right: ▲ | ●

Bottom-left: □ | ◆

Bottom-right: ?

Across each row, the right symbol of a cell becomes the left symbol of the next cell, and the new right symbol is the filled form of the previous cell's left symbol. Which cell completes the matrix?

- A) □ | ◆
- B) ◆ | ■
- C) ■ | ◆
- D) ◆ | □

164. Complete the 2×2 matrix.

Top-left: ↑ ○

Top-right: ○ →

Bottom-left: ↓ □

Bottom-right: ?

Across a row, the arrow moves from before the shape to after it and rotates 90° clockwise. The shape stays the same. Which cell completes the matrix?

- A) ← □
- B) □ ↓
- C) ↑ □
- D) □ ←

165. Complete the 2×2 matrix.

Top-left: ○ + □

Top-right: ○ □ +

Bottom-left: △ + ◇

Bottom-right: ?

Across each row, the middle symbol moves to the far right while the other two remain in order. Which cell completes the matrix?

- A) + △ ◇
- B) ◇ △ +
- C) △ ◇ +
- D) △ + ◇

166. Complete the 2×2 matrix.

Top-left: Top ○; Bottom ■

Top-right: Top ■; Bottom ○

Bottom-left: Top △; Bottom ◆

Bottom-right: ?

Across each row, the two symbols exchange vertical positions. Which cell completes the matrix?

- A) Top ◆; Bottom △
- B) Top △; Bottom ◆
- C) Top ▲; Bottom ◇
- D) Top ◇; Bottom ▲

167. A square sheet is folded along the diagonal from upper-left to lower-right so the lower-left half lies over the upper-right half. A triangular notch is cut from the folded edge halfway between the upper-left and lower-right corners, not at either endpoint. What appears after the sheet is fully unfolded?

- A) One triangular notch on the outer left edge
- B) Two triangular notches mirrored across the other diagonal
- C) Two triangular cutouts mirrored across the fold diagonal
- D) Four triangular cutouts arranged around the center

168. A rectangular sheet is folded from bottom to top along the horizontal centerline. The folded rectangle is then folded from right to left along its vertical centerline. One hole is punched exactly at the upper-left corner of the final folded rectangle. Where will the holes appear after full unfolding?

- A) At the centers of the four quarters
- B) At four locations meeting at the original sheet's center
- C) At the midpoint of the top and bottom edges only
- D) At the four outer corners of the original sheet

169. A square sheet is folded from left to right along the vertical centerline. One circular hole is punched on the top edge of the folded rectangle, halfway between its left and right edges. After unfolding, what is the result?

- A) One hole at the midpoint of the square's top edge
- B) Two holes on the bottom edge
- C) Two holes on the top edge, equally spaced from the vertical centerline
- D) Four holes, one near each corner

170. A rectangular sheet that is twice as wide as it is tall is folded from left to right along the vertical centerline, forming a square. The square is then folded along the diagonal from its upper-left corner to its lower-right corner. A hole is punched inside the final folded shape at a point not on either fold. What must be true after full unfolding?

- A) Exactly two holes appear, both above the horizontal centerline.
- B) Exactly eight holes appear around the outer edge.
- C) Exactly six holes appear along one diagonal.
- D) Exactly four holes appear, related by the two fold reflections.

171. The three groups below share a rule.

○ ■ ○

□ ▲ □

◇ ★ ◇

Which choice follows the same rule?

A) △ ● □

B) ☆ ◆ ☆

C) ○ ○ ●

D) ■ ○ ■

172. The three groups below share a rule.

↑ ○ ↓

→ □ ←

↓ △ ↑

Which choice follows the same rule?

A) ← ◇ ↓

B) → ◇ ←

C) ↓ ○ ←

D) ↑ ☆ →

173. The three groups below share a rule.

○ | ●

□ - ■

△ / ▲

Which choice follows the same rule?

A) ★ | ☆

B) ◆ / ◇

C) ◇ - ◆

D) ☆ × ★

174. The three arrangements below share a rule.

Top: ○; Center: +; Bottom: ○

Top: □; Center: ×; Bottom: □

Top: △; Center: |; Bottom: △

Which choice follows the same rule?

A) Top: -; Center: ◇; Bottom: -

B) Top: ◇; Center: ◇; Bottom: -

C) Top: ◆; Center: -; Bottom: ◇

D) Top: ◇; Center: -; Bottom: ◇

175. Complete the 2 × 2 matrix.

Top-left: ○ ○ □

Top-right: □

Bottom-left: △ ◇ ◇

Bottom-right: ?

Across each row, the matching pair is removed and the unpaired shape remains. Which cell completes the matrix?

A) △ △

B) ◇

C) △ ◇

D) △

176. Complete the 2×2 matrix.

Top-left: Top-left ↑; Bottom-right ○

Top-right: Top-right →; Bottom-left ○

Bottom-left: Bottom-left ←; Top-right □

Bottom-right: ?

Across each row, the arrow moves one corner clockwise while the shape moves one corner counterclockwise. The arrow also rotates 90° clockwise. Which cell completes the matrix?

A) Top-left ↓; Bottom-right □

B) Top-right ↑; Bottom-left □

C) Top-left ↑; Bottom-right □

D) Top-left ←; Bottom-right □

177. Complete the 2×2 matrix.

Top-left: ○

Top-right: ○ ○

Bottom-left: ▲ ▲

Bottom-right: ?

Across each row, one matching shape is added. Down each column, open circles become filled triangles and the count doubles. Which cell completes the matrix?

A) ▲ ▲ ▲ ▲

B) ▲ ▲ ▲

C) △ △ △ △

D) ▲ ▲

178. Complete the 2×2 matrix.

Top-left: ○ inside □

Top-right: □ inside ○

Bottom-left: △ inside ◇

Bottom-right: ?

Across each row, the inner and outer shapes exchange places without changing fill. Which cell completes the matrix?

A) △ beside ◇

B) ◇ inside △

C) ▲ inside ◆

D) △ inside ◇

179. Complete the 3×3 matrix.

Row 1: ○ □ | □ △ | ○ △

Row 2: ▲ ◆ | ◆ ★ | ▲ ★

Row 3: Cell 1: + ×; Cell 2: × |; Cell 3: ?

In each row, the symbol appearing in both of the first two cells is removed; the third cell contains the two remaining symbols in order. Which cell completes the matrix?

A) | +

B) + |

C) + × |

D) × |

180. A square sheet is folded from left to right along the vertical centerline and then from bottom to top along the horizontal centerline. The resulting small square is folded along its diagonal from upper-left to lower-right. One hole is punched inside the final folded shape, away from folds and edges. After full unfolding, what appears?

- A) Six holes arranged along the diagonal
- B) Four holes related only by the centerline folds
- C) Eight holes related by three fold reflections
- D) Two holes on opposite sides of the diagonal

181. A circular sheet is folded in half from left to right along a vertical diameter. A small semicircular notch is cut from the curved outer edge of the folded sheet, above the horizontal center. What appears after unfolding?

- A) One circular hole at the center
- B) Two matching notches on the curved edge, mirrored across the vertical diameter
- C) Two holes on the vertical diameter
- D) Four matching notches equally spaced around the edge

182. A rectangular sheet is folded from top to bottom along the horizontal centerline. A hole is punched exactly on the folded sheet's vertical centerline and below the fold, not on an outer edge. What appears after unfolding?

- A) Two holes on the vertical centerline, mirrored across the horizontal centerline
- B) Two holes on the horizontal centerline
- C) One hole above the horizontal centerline
- D) Four holes mirrored across both centerlines

183. A square sheet is folded along the diagonal from upper-right to lower-left. The resulting right isosceles triangle is folded along its line of symmetry so that its two acute corners meet. A hole is punched inside the final folded shape, away from folds and edges. How many holes appear after full unfolding?

- A) 2
- B) 3
- C) 6
- D) 4

184. The three groups below share a rule.

○ → ■
 ▲ ← □
 ◇ → ★

In each group, the arrow points toward the filled shape. Which choice follows the same rule?

- A) ◆ ← ☆
- B) ☆ ← ◆
- C) ◆ → ☆
- D) ☆ → ◇

185. The three groups below share a rule.

↑ → ↓
→ ↓ ←
↓ ← ↑

Which choice follows the same rule?

- A) ← ↑ →
- B) ← ↓ →
- C) ↑ ← ↓
- D) → ↑ ←

186. The three arrangements below share a rule.

Top-left: ○; Top-right: ■; Bottom: □

Top-left: △; Top-right: ◆; Bottom: ◇

Top-left: ☆; Top-right: ●; Bottom: ○

Which choice follows the same rule?

- A) Top-left: □; Top-right: ■; Bottom: △
- B) Top-left: ◆; Top-right: ▲; Bottom: △
- C) Top-left: □; Top-right: ▲; Bottom: △
- D) Top-left: □; Top-right: △; Bottom: ▲

187. The three arrangements below share a rule.

Top: ○ □; Bottom: ■ ●

Top: △ ◇; Bottom: ◆ ▲

Top: ☆ ○; Bottom: ● ★

Which choice follows the same rule?

- A) Top: □ ◇; Bottom: ■ ◆
- B) Top: □ ◇; Bottom: ◆ ■
- C) Top: ■ ◇; Bottom: ◆ □
- D) Top: □ ◇; Bottom: ◇ □

188. Complete the 3 × 3 matrix.

Row 1: Cell 1: ○ □; Cell 2: □ △; Cell 3: ○ □ △

Row 2: Cell 1: ▲ ◆; Cell 2: ◆ ★; Cell 3: ▲ ◆ ★

Row 3: Cell 1: + ×; Cell 2: × |; Cell 3: ?

In each row, the third cell contains every different symbol appearing in the first two cells, with the shared symbol written only once. Which cell completes the matrix?

- A) | × +
- B) + |
- C) × × |
- D) + × |

189. Complete the 2 × 2 matrix.

Top-left: ○ | □

Top-right: □ | △

Bottom-left: △ | ◇

Bottom-right: ?

Across each row, each symbol advances one place in this cycle: ○, □, △, ◇, ☆. Which cell completes the matrix?

- A) ◇ | ☆
- B) △ | ☆
- C) ☆ | ◇
- D) ◇ | △

190. Complete the 2×2 matrix.

Top-left: ○ ○ ○ ○

Top-right: ● ● ●

Bottom-left: □ □ □

Bottom-right: ?

Across each row, one shape is removed and all remaining shapes change from open to filled. Which cell completes the matrix?

A) □ □

B) ■ ■ ■

C) ■ ■

D) □ □ □ □

191. Complete the 2×2 matrix.

Top-left: ○ + □

Top-right: ○ × □

Bottom-left: △ × ◇

Bottom-right: ?

Across each row, the outer shapes remain fixed while the center line symbol rotates 45° clockwise. Which cell completes the matrix?

A) △ + ◇

B) ◇ - △

C) △ | ◇

D) △ - ◇

192. A rectangular sheet is divided conceptually into three equal vertical panels. The left panel is folded rightward over the center panel, and the right panel is then folded leftward over the center panel. One hole is punched inside the final stack, away from folds and edges. What appears after full unfolding?

A) Two holes mirrored across one fold

B) Four holes arranged around the center

C) Three holes, one in each vertical panel at matching heights

D) Six holes in two rows

193. A square sheet is folded from bottom to top along the horizontal centerline. A V-shaped notch is cut from the folded sheet's left outer edge, above the fold. What appears after unfolding?

A) One notch on the left edge

B) Two matching notches on the left edge, mirrored across the horizontal centerline

C) Four notches, one on each edge

D) Two notches on the top and bottom edges

194. A square sheet is folded along the diagonal from lower-left to upper-right. One hole is punched exactly on the folded diagonal, away from the corners. What appears after unfolding?

A) One hole located on the diagonal fold line

B) Two holes mirrored across the diagonal

C) Four holes around the center

D) One hole on an outer edge

195. A rectangular sheet is folded from left to right along the vertical centerline. Two holes are punched inside the folded rectangle at different heights and different distances from the fold, with neither hole on a fold or edge. How many holes appear after unfolding?

- A) 2
- B) 4
- C) 3
- D) 8

196. The three arrangements below share a rule.

Top-left: /; Top-right: \; Bottom: ×

Top-left: |; Top-right: -; Bottom: +

Top-left: \; Top-right: /; Bottom: ×

The bottom symbol combines the two line directions above it. Which choice follows the same rule?

- A) Top-left: /; Top-right: |; Bottom: +
- B) Top-left: |; Top-right: |; Bottom: -
- C) Top-left: -; Top-right: /; Bottom: ×
- D) Top-left: -; Top-right: |; Bottom: +

197. The three groups below share a rule.

○ ● □ ■

△ ▲ ◇ ◆

☆ ★ ○ ●

Which choice follows the same rule?

- A) □ ■ ◆ ◇
- B) □ ■ △ ▲
- C) ◇ ◆ ▲ △
- D) □ △ ■ ▲

198. The three arrangements below share a rule.

Top-left: ○; Top-right: ■; Bottom-left: □; Bottom-right: ●

Top-left: △; Top-right: ◆; Bottom-left: ◇; Bottom-right: ▲

Top-left: ☆; Top-right: ●; Bottom-left: ○; Bottom-right: ★

In each arrangement, the left column contains open shapes, the right column contains their filled counterparts, and each counterpart lies diagonally opposite its open form.

Which choice follows the same rule?

- A) Top-left: □; Top-right: ▲; Bottom-left: △; Bottom-right: ■
- B) Top-left: ◆; Top-right: ▲; Bottom-left: △; Bottom-right: ◇
- C) Top-left: □; Top-right: △; Bottom-left: ▲; Bottom-right: ■
- D) Top-left: □; Top-right: ▲; Bottom-left: ○; Bottom-right: ■

199. The three groups below share a rule.

↑ ↑ ↓

→ → ←

↓ ↓ ↑

Which choice follows the same rule?

- A) ← ← →
- B) ← → →
- C) ↑ ↓ ↑
- D) → ← ←

200. The three groups below share a rule.

○ ■ □

△ ◆ ◇

☆ ● ○

In each group, the center is the filled counterpart of the open shape on the right; the left symbol is different. Which choice follows the same rule?

A) ◇ ▲ □

B) + ▲ △

C) + △ ▲

D) ◆ ◇ △

Answer Key

Question numbers continue across all five sections.

Reading

1. D	2. C	3. D	4. D	5. B	6. D	7. A	8. A
9. C	10. D	11. A	12. C	13. A	14. D	15. A	16. A
17. B	18. D	19. B	20. A	21. B	22. C	23. B	24. C
25. A	26. B	27. D	28. C	29. C	30. B	31. B	32. C
33. D	34. D	35. A	36. B	37. C	38. B	39. C	40. B
41. A	42. B	43. A	44. A	45. D	46. A	47. D	48. C
49. A	50. C						

Written Expression

51. A	52. C	53. D	54. A	55. B	56. B	57. A	58. C
59. A	60. B	61. A	62. A	63. B	64. B	65. D	66. D
67. B	68. D	69. C	70. D	71. C	72. D	73. B	74. C
75. D	76. C	77. C	78. A	79. B	80. D	81. B	82. B
83. C	84. C	85. A	86. A	87. C	88. A	89. C	90. C
91. D	92. C	93. A	94. B	95. D	96. B	97. A	98. B
99. D	100. D						

Mathematics

101. C	102. B	103. A	104. D	105. A	106. C	107. D	108. A
109. B	110. A	111. B	112. A	113. C	114. C	115. D	116. A
117. B	118. C	119. C	120. D	121. D	122. B	123. B	124. C
125. D	126. C	127. B	128. D	129. B	130. C	131. A	132. D
133. C	134. A	135. B	136. D	137. D	138. A	139. C	140. B
141. D	142. B	143. A	144. D	145. B	146. A	147. D	148. A
149. C	150. C						

Ability

151. D	152. D	153. B	154. D	155. C	156. D	157. C	158. B
159. C	160. D	161. A	162. A	163. B	164. D	165. C	166. A
167. C	168. D	169. C	170. D	171. B	172. B	173. C	174. D
175. D	176. C	177. A	178. B	179. B	180. C	181. B	182. A
183. D	184. A	185. A	186. C	187. B	188. D	189. A	190. C
191. A	192. C	193. B	194. A	195. B	196. D	197. B	198. A

199. A	200. B						
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Detailed Explanations

Reading

1. TACHS-R8-001 - Answer D: Opening it would release heat.
The uncle explains that repeatedly opening the oven would allow heat to escape.
2. TACHS-R8-002 - Answer C: The oven is already hot from the first batch.
The passage states that the oven retained heat between batches, shortening the second baking time.
3. TACHS-R8-003 - Answer D: Combining measurements with direct observation leads to better judgment.
Mina learns to connect elapsed time with visible and sensory signs rather than relying on one signal alone.
4. TACHS-R8-004 - Answer D: regarded or considered.
Mina no longer regarded the timer's ring as the only useful evidence.
5. TACHS-R8-005 - Answer B: Small pockets of air
Tiny bumps can support the droplet while small air pockets remain beneath it.
6. TACHS-R8-006 - Answer D: To clarify the limits of the effect
The statement prevents readers from interpreting a useful effect as complete protection.
7. TACHS-R8-007 - Answer A: A surface effect is described, explained, and connected to possible applications and limits.
The passage moves from observation to mechanism, benefits, limitations, and engineering applications.
8. TACHS-R8-008 - Answer A: A durable microscopic texture combined with water-repelling chemistry
The passage emphasizes that effective repellence requires both suitable texture and surface chemistry.
9. TACHS-R8-009 - Answer C: Repeated contact with carts or equipment
The passage identifies carts or equipment as a possible cause of low scratches.
10. TACHS-R8-010 - Answer D: care to avoid unsupported conclusions.
Researchers must interpret physical evidence carefully and consider more than one explanation.
11. TACHS-R8-011 - Answer A: Doorway design, wear, and alteration can provide evidence when interpreted with other sources.
The passage describes several kinds of doorway evidence and stresses careful comparison with other records.
12. TACHS-R8-012 - Answer C: It can be adapted for a new use while preserving evidence of the old structure.
The passage explains that an old loading opening may become a large window while retaining evidence of the warehouse.
13. TACHS-R8-013 - Answer A: curious but careful about the limits of evidence.
The author encourages investigation while repeatedly warning against conclusions that exceed the evidence.
14. TACHS-R8-014 - Answer D: Its leg catches on a strip of tape.
The passage states that the chair's metal leg catches on the taped court boundary.
15. TACHS-R8-015 - Answer A: It reminds her of a way to judge balance by examining surrounding space.
Seeing Mr. Imani recalls his earlier method of comparing empty spaces to assess an arrangement.
16. TACHS-R8-016 - Answer A: A solution may emerge when a person considers the whole arrangement instead of one part.
Mara succeeds by shifting the row and camera rather than treating the chair as an isolated problem.
17. TACHS-R8-017 - Answer B: show the pressure Mara feels before she pauses and rethinks the problem.
Her embarrassment heightens the moment of difficulty that leads to her more thoughtful approach.
18. TACHS-R8-018 - Answer D: Surface water generally moves faster than water slowed near the banks and bottom.
Friction slows other parts of the stream, so the surface measurement requires adjustment.
19. TACHS-R8-019 - Answer B: done as an early step.
The measurement is preliminary because it is taken before a team chooses where to gather detailed data.
20. TACHS-R8-020 - Answer A: It explains a method, identifies sources of error, and describes appropriate uses.
The three paragraphs follow that sequence from procedure to limitations to applications.
21. TACHS-R8-021 - Answer B: A simple measurement can be useful when its limitations are understood.
The passage values the method while emphasizing correction factors, repeated trials, and appropriate uses.
22. TACHS-R8-022 - Answer C: To keep printed words from disappearing into the binding
The first paragraph directly explains the practical need for inner margin space.
23. TACHS-R8-023 - Answer B: closely packed.
Dense writing appears cramped when little blank space surrounds text that is closely packed.

24. TACHS-R8-024 - Answer C: Margins have long supported both the practical use and the visual design of pages. The passage traces how margins protect readability and shape appearance across old and modern formats.
25. TACHS-R8-025 - Answer A: To demonstrate that the purposes of margins continue across changing formats. The comparison shows that different formats still require spacing for readability and visual character.
26. TACHS-R8-026 - Answer B: It controls the ventilation system. The circuit map identifies the unlabeled switch as the control for ventilation equipment.
27. TACHS-R8-027 - Answer D: She hopes future operators can verify controls instead of guessing. Nina labels the reset and protects the map so later users can identify the correct control.
28. TACHS-R8-028 - Answer C: Careful investigation is safer and more effective than an appealing guess. Nina avoids the tempting switch, consults reliable information, and finds the proper reset.
29. TACHS-R8-029 - Answer C: return to working condition. In context, restoring the light means making the unresponsive spotlight work again.
30. TACHS-R8-030 - Answer B: It moves backward in a circle and tosses grains outward. The first paragraph gives this action as the beginning of pit construction.
31. TACHS-R8-031 - Answer B: They hold together instead of sliding easily. The passage explains that damp grains cling together and do not flow readily.
32. TACHS-R8-032 - Answer C: provide evidence that grain properties affect how well the trap functions. The test results connect easily shifted material with steeper, more effective pits.
33. TACHS-R8-033 - Answer D: It explains the broader principle that makes the constructed slope effective. The final paragraph connects the larva's behavior to the ordinary collapse point of loose grains.
34. TACHS-R8-034 - Answer D: The position and type of a knot. The first paragraph states that both features helped distinguish values in the place-value system.
35. TACHS-R8-035 - Answer A: able to be touched or physically handled. The cords made values physical objects that users could handle, group, and inspect.
36. TACHS-R8-036 - Answer B: Khipus organized information through physical features, although some features remain uncertain. This choice includes both their evidence-supported numerical role and the unresolved meaning of other features.
37. TACHS-R8-037 - Answer C: To show that researchers must combine several kinds of evidence. Researchers study construction, archaeological setting, and written accounts because no clue is sufficient alone.
38. TACHS-R8-038 - Answer B: Respectful of the evidence while cautious about uncertain interpretations. The author explains khipus' achievements while distinguishing supported conclusions from possibilities.
39. TACHS-R8-039 - Answer C: Tessa's slippery hand catches on the rough patch. Tessa's experience reveals that the rough area provides a useful grip.
40. TACHS-R8-040 - Answer B: To make the useful texture appear deliberate and consistent. Jonah repeats and evens the texture after recognizing its practical value.
41. TACHS-R8-041 - Answer A: examine an apparent flaw before deciding how to correct it. Jonah tests the marks and discovers that they improve the handle's function.
42. TACHS-R8-042 - Answer B: done by design. Jonah refines and repeats the marks so they look purposely included.
43. TACHS-R8-043 - Answer A: Near the spider's leg joints. The second paragraph directly locates the vibration-sensitive organs near the leg joints.
44. TACHS-R8-044 - Answer A: Spiders use varied vibration patterns in maintained webs to gather information. The passage explains how web signals differ, how spiders interpret them, and why web condition matters.
45. TACHS-R8-045 - Answer D: carry from one place to another. A damaged strand may carry movement differently through the web.
46. TACHS-R8-046 - Answer A: To show that a spider must distinguish useful signals from other movement. These examples represent background vibrations that differ from the signal of trapped prey.
47. TACHS-R8-047 - Answer D: Parchment was costly and could be reused. The opening paragraph identifies expense as the reason a sheet might carry a new text.
48. TACHS-R8-048 - Answer C: Overlapping marks, missing letters, and damage can make readings uncertain. Those conditions may support more than one interpretation of a faded word.

49. TACHS-R8-049 - Answer A: Reused parchment can preserve layered evidence that modern methods help scholars interpret.

The passage connects palimpsests' multiple texts with imaging and careful scholarly analysis.

50. TACHS-R8-050 - Answer C: It defines a type of page, explains how it is studied, and considers what it reveals.

The paragraphs move from definition to recovery methods and finally to the page's historical meaning.

Written Expression

51. TACHS-WE8-001 - Answer A: Sentence 5

The pile's location behind the shed does not explain measurement, temperature change, or response actions.

52. TACHS-WE8-002 - Answer C: Twice each week, students insert the same thermometer to the same depth, wait one minute, and record the compost's internal temperature.

Choice C combines the measurement schedule and procedure with clear parallel verbs.

53. TACHS-WE8-003 - Answer D: mixing the pile or adding water

The replacement names the specific actions described earlier and removes vague wording.

54. TACHS-WE8-004 - Answer A: Microbial activity often decreases as available material changes.

The sentence explains why the pile may later cool and prepares for the response described in sentence 6.

55. TACHS-WE8-005 - Answer B: Each frame has a label with its identification number.

The singular subject "Each frame" requires the singular verb "has."

56. TACHS-WE8-006 - Answer B: Before and after each loan, borrowers would complete a condition form for the framed work identified by its numbered label.

Choice B connects the condition record, loan timing, and identification label in one complete sentence.

57. TACHS-WE8-007 - Answer A: Two people would carry large pieces, and borrowers would keep all artwork out of direct sunlight.

Choice A states the two handling rules directly and without redundancy.

58. TACHS-WE8-008 - Answer C: Student artwork can continue serving an audience even when it is not in the main gallery.

The sentence introduces the value of sharing stored art and leads naturally to the lending proposal.

59. TACHS-WE8-009 - Answer A: The number of applicants has increased.

The singular subject phrase "The number" correctly takes "has."

60. TACHS-WE8-010 - Answer B: After the meeting, we reviewed the notes.

A comma correctly follows the introductory prepositional phrase.

61. TACHS-WE8-011 - Answer A: This route is safer than the old one.

"Safer than" is the clear and correct comparative form for two routes.

62. TACHS-WE8-012 - Answer A: Line 4

Line 4 requires "were" because the subject "seats" is plural.

63. TACHS-WE8-013 - Answer B: Line 1

Line 1 requires the object pronoun "me" after the preposition "to."

64. TACHS-WE8-014 - Answer B: Sentence 7

The custodian's arrival time is unrelated to testing the scale or improving measurement accuracy.

65. TACHS-WE8-015 - Answer D: Next, we weighed the container five times and recorded every result.

This active construction states both actions directly without wordiness or unclear references.

66. TACHS-WE8-016 - Answer D: Because the readings became consistent on the tile, we learned that a measuring tool's surroundings can influence its results.

The subordinate clause clearly connects the observed result with the club's conclusion.

67. TACHS-WE8-017 - Answer B: After sentence 1

The added sentence identifies the controlled object before the passage describes changes to the scale's surroundings.

68. TACHS-WE8-018 - Answer D: We created three small desk cards: green meant "working independently," yellow meant "question when available," and red meant "help needed now."

A colon correctly introduces the explanatory list, and commas separate its three parallel items.

69. TACHS-WE8-019 - Answer C: As students learned when to use each signal, they reserved the red card for problems that stopped their progress.

The revision precisely explains both the cause of the change and the red card's intended use.

70. TACHS-WE8-020 - Answer D: The cards were laminated, which made them durable enough for daily use.

The relative clause correctly joins the related ideas and eliminates the comma splice.

71. TACHS-WE8-021 - Answer C: A simple visual system can support both concentration and timely assistance.

This conclusion restates the passage's broader lesson without introducing an unrelated detail.

72. TACHS-WE8-022 - Answer D: The second sample is more stable than the first sample.

"More stable" correctly compares two samples without a double comparative or superlative.

73. TACHS-WE8-023 - Answer B: The librarian asked Noah and me to close the windows.
"Me" is the correct objective pronoun as part of the compound object of "asked."
74. TACHS-WE8-024 - Answer C: Line 4
The singular subject "Neither" requires "is," not "are."
75. TACHS-WE8-025 - Answer D: Line 2
The past narrative requires "began" to remain consistent with "arrived," "was seated," "answered," and "ended."
76. TACHS-WE8-026 - Answer C: Line 4
The verb "handled" must be modified by the adverb "carefully," not the adjective "careful."
77. TACHS-WE8-027 - Answer C: Each shade covered the same area of glass during its trial.
This detail describes a controlled condition that makes comparison among the materials fair.
78. TACHS-WE8-028 - Answer A: We also observed that the silver surface reflected much of the incoming light.
The revision is complete, direct, and explains a relevant property without a misplaced modifier.
79. TACHS-WE8-029 - Answer B: Because the fabric limited warming and remained easy to store, we recommended a two-week trial in another classroom.
The revision names both findings and clearly connects them to the recommendation.
80. TACHS-WE8-030 - Answer D: We compared the changes recorded for all three materials.
This sentence provides the comparison step needed before reporting which material performed best.
81. TACHS-WE8-031 - Answer B: Volunteers collected the holders at the exit, wiped them clean, and returned them to a labeled box.
All three coordinated verbs correctly use the past tense.
82. TACHS-WE8-032 - Answer B: Although the welcome committee printed new paper name badges for every event, most were discarded after only a few hours.
The dependent clause and main clause form a complete sentence that clearly shows contrast.
83. TACHS-WE8-033 - Answer C: Four holders were missing, so the committee replaced them and reused the remaining name cards.
This version preserves the information in a direct sentence without unnecessary wording.
84. TACHS-WE8-034 - Answer C: The guide asked us to label the samples, record the weights, and store the boxes.
The three coordinated verbs use matching base forms and share the phrase "asked us to."
85. TACHS-WE8-035 - Answer A: The bicycles were moved away from the gate because they blocked the entrance.
The plural pronoun "they" clearly refers to "bicycles."
86. TACHS-WE8-036 - Answer A: Before the bus arrived, we checked the list one final time.
A comma correctly separates the introductory dependent clause from the independent clause.
87. TACHS-WE8-037 - Answer C: Line 3
The count noun "measuring cups" requires "fewer," not "less."
88. TACHS-WE8-038 - Answer A: Line 1
The introductory phrase illogically modifies "banner"; Elena, not the banner, was walking.
89. TACHS-WE8-039 - Answer C: Sentence 6
The tray's color does not contribute to the process, results, or purpose of the lending system.
90. TACHS-WE8-040 - Answer C: We assigned numbers to the magnifiers and attached matching tags to their cases.
The plural nouns and pronoun correctly refer to multiple magnifiers and cases without unnecessary apostrophes.
91. TACHS-WE8-041 - Answer D: In the first month, the system prevented losses and helped the club identify minor damage promptly.
The revision states both outcomes precisely and avoids vague wording.
92. TACHS-WE8-042 - Answer C: After returning students placed the magnifiers in a shallow tray, a club monitor checked the lenses and returned each tool to its numbered case.
Subordination clearly connects the two actions without a comma splice or misplaced modifier.
93. TACHS-WE8-043 - Answer A: Groups entered their names in fifteen-minute blocks, erasing each entry after finishing.
The participial phrase correctly joins the related actions and removes the comma splice.
94. TACHS-WE8-044 - Answer B: The council compared hallway observations from before and after the schedule was posted.
This sentence introduces the comparison that supports the results reported next.
95. TACHS-WE8-045 - Answer D: Any unreserved group could use an open time block.
The revision expresses the complete idea directly and eliminates repeated wording.

96. TACHS-WE8-046 - Answer B: The nearly complete model stood on the shelf.
The adverb “nearly” correctly modifies the adjective “complete.”
97. TACHS-WE8-047 - Answer A: The committee has chosen three finalists.
The present perfect construction requires the past participle “chosen.”
98. TACHS-WE8-048 - Answer B: Line 2
“More clearer” is an incorrect double comparative; “clearer” is sufficient.
99. TACHS-WE8-049 - Answer D: Line 4
The intransitive past-tense form is “lay,” so the envelope “lay there.”
100. TACHS-WE8-050 - Answer D: Line 2
The singular subject “One” requires the verb “was.”

Mathematics

101. TACHS-M8-001 - Answer C: 8

$3\frac{1}{2} = \frac{7}{2}$ and $2\frac{2}{7} = \frac{16}{7}$; their product is 8.

102. TACHS-M8-002 - Answer B: 7

Subtract 5 to get $8x = 56$, so $x = 7$.

103. TACHS-M8-003 - Answer A: 10

The range is $14 - 4 = 10$.

104. TACHS-M8-004 - Answer D: 110°

Pentagon angles total 540° ; the known angles total 430° , so $x = 110^\circ$.

105. TACHS-M8-005 - Answer A: 0.4 km

$2.4 \div 6 = 0.4$ kilometer.

106. TACHS-M8-006 - Answer C: 80

$1.25 \times 64 = 80$.

107. TACHS-M8-007 - Answer D: Divide by 3.

Each term is the preceding term divided by 3.

108. TACHS-M8-008 - Answer A: $\frac{3}{5}$

There are 6 blue-or-green beads among 10 total, giving $\frac{3}{5}$.

109. TACHS-M8-009 - Answer B: 44 units

The side length is $\sqrt{121} = 11$, so the perimeter is 44.

110. TACHS-M8-010 - Answer A: 126

The rate is 7 bottles per minute, and $7 \times 18 = 126$.

111. TACHS-M8-011 - Answer B: 18

The greatest shared factor of 54 and 90 is 18.

112. TACHS-M8-012 - Answer A: $4n - 7 < 21$

Four times n is $4n$; decreased by 7 gives $4n - 7$, and "less than" means $<$.

113. TACHS-M8-013 - Answer C: 7

Multiplying by 0.4 and dividing by 0.05 multiplies the original number by 8, so $56 \div 8 = 7$.

114. TACHS-M8-014 - Answer C: 20

The total must be $5 \times 18 = 90$; the known values total 70, leaving 20.

115. TACHS-M8-015 - Answer D: 66°

If the smaller angle is x , then $x + (2x - 18) = 180$, so $x = 66$.

116. TACHS-M8-016 - Answer A: 1,750 mL

Convert 2.4 liters to 2,400 milliliters, then calculate $2,400 - 650 = 1,750$.

117. TACHS-M8-017 - Answer B: $\frac{13}{20}$

$\frac{3}{5} \div \frac{3}{2} = \frac{2}{5}$, and $\frac{2}{5} + \frac{1}{4} = \frac{8}{20} + \frac{5}{20} = \frac{13}{20}$.

118. TACHS-M8-018 - Answer C: 13

Squaring -4 gives 16, and $16 - 3 = 13$.

119. TACHS-M8-019 - Answer C: $\frac{1}{22}$

The probability is $\frac{3}{12} \times \frac{2}{11} = \frac{6}{132} = \frac{1}{22}$.

120. TACHS-M8-020 - Answer D: 225 square units

The hypotenuse satisfies $9^2 + 12^2 = 225$, which is also the area of its constructed square.

121. TACHS-M8-021 - Answer D: The groups have the same mean, but Group P has a smaller range.

Both totals are 50, while the ranges are 7 for P and 9 for Q.

122. TACHS-M8-022 - Answer B: 18 cm^2

The border leaves an inner rectangle measuring 6 centimeters by 3 centimeters, with area 18 square centimeters.

123. TACHS-M8-023 - Answer B: 120

Since 8 and 15 share no prime factors, their least common multiple is $8 \times 15 = 120$.

124. TACHS-M8-024 - Answer C: 49

Each term is twice the preceding term minus 1, so $2 \times 25 - 1 = 49$.

125. TACHS-M8-025 - Answer D: 15
The exterior angles total 360° , so the polygon has $360 \div 24 = 15$ sides.
126. TACHS-M8-026 - Answer C: $1\frac{3}{8}$
Convert 2.75 to $2\frac{3}{4}$, then $2\frac{3}{4} - 1\frac{3}{8} = 1\frac{3}{8}$.
127. TACHS-M8-027 - Answer B: 13 cm
If the base is b , then $b + 2(b + 4) = 47$, so $3b = 39$ and $b = 13$.
128. TACHS-M8-028 - Answer D: $\frac{5}{8}$
The favorable numbers are 1, 2, 4, 7, and 8, giving 5 favorable outcomes out of 8.
129. TACHS-M8-029 - Answer B: 80°
The nine ratio parts total 180° , so each part is 20° and the largest angle is $4 \times 20^\circ$.
130. TACHS-M8-030 - Answer C: 1,350 m
Convert 3.6 kilometers to 3,600 meters and subtract 2,250 to get 1,350 meters.
131. TACHS-M8-031 - Answer A: \$42
A 25% discount leaves \$48, and the \$6 coupon reduces the price to \$42.
132. TACHS-M8-032 - Answer D: 6
Expanding gives $10x - 15 = 4x + 21$, so $6x = 36$ and $x = 6$.
133. TACHS-M8-033 - Answer C: 18
Adding 18 keeps 11 as the middle value and gives a range of $18 - 6 = 12$.
134. TACHS-M8-034 - Answer A: 81π cm²
From $2\pi r = 18\pi$, $r = 9$; therefore, $\pi r^2 = 81\pi$ square centimeters.
135. TACHS-M8-035 - Answer B: 49
The rate is $14 \div 8$ containers per minute, so $28 \times 14 \div 8 = 49$.
136. TACHS-M8-036 - Answer D: $\frac{7}{8} - \frac{1}{4}$
The values are $\frac{1}{2}$, $\frac{5}{8}$, $\frac{1}{2}$, and 0.55; $\frac{5}{8}$ is greatest.
137. TACHS-M8-037 - Answer D: $y = 2x + 1$
The slope is 2, and substituting (2, 5) gives the y-intercept 1.
138. TACHS-M8-038 - Answer A: -14
Compute $-18 \div 3 = -6$ and $4(-2) = -8$; then $-6 + (-8) = -14$.
139. TACHS-M8-039 - Answer C: $\frac{2}{3}$
Eight of the twelve clips are not blue, so the probability is $\frac{8}{12} = \frac{2}{3}$.
140. TACHS-M8-040 - Answer B: 40 units
The dimensions are 8 and 12 because $8 \times 12 = 96$; the perimeter is $2(8 + 12) = 40$.
141. TACHS-M8-041 - Answer D: 7
Solving gives $x \leq 6$, so 7 is the listed value that does not satisfy the inequality.
142. TACHS-M8-042 - Answer B: 58.5 miles
Multiply 3.25 inches by 18 miles per inch to obtain 58.5 miles.
143. TACHS-M8-043 - Answer A: 56
Since $56 = 2^3 \times 7$, all of its prime factors are available in the original number.
144. TACHS-M8-044 - Answer D: 24
The six-score total is 84 and the remaining total is 60, so the removed score is 24.
145. TACHS-M8-045 - Answer B: 136 square units
Surface area is $2(3 \times 4 + 3 \times 8 + 4 \times 8) = 2(68) = 136$.
146. TACHS-M8-046 - Answer A: $\frac{1}{9}$
Dividing by 3 each time gives terms five through seven as 1, $\frac{1}{3}$, and $\frac{1}{9}$.
147. TACHS-M8-047 - Answer D: 14 miles per hour
The cyclist travels 42 miles in 3 hours, so the overall average speed is $42 \div 3 = 14$.
148. TACHS-M8-048 - Answer A: 65
If 1.2 times the original is 78, then the original is $78 \div 1.2 = 65$.
149. TACHS-M8-049 - Answer C: 9
Adding 9 gives seven ordered values with 9 in the fourth position, so the median becomes 9.

150. TACHS-M8-050 - Answer C: 12

The exterior angle is $180^\circ - 150^\circ = 30^\circ$, and $360^\circ \div 30^\circ = 12$ sides.

Ability

151. TACHS-A8-001 - Answer D: $\triangle$ bottom-right; $\diamond$ top-left

One clockwise corner move sends $\triangle$ to bottom-right and $\diamond$ to top-left.

152. TACHS-A8-002 - Answer D: $\triangle$ inside $\square$

The open triangle remains unchanged and gains an outer square.

153. TACHS-A8-003 - Answer B: $\square\square\times$

The square is duplicated and $+$ changes to $\times$.

154. TACHS-A8-004 - Answer D: $/\blacksquare$

A left-right reflection changes $\backslash$ to $/$, and open $\square$ becomes filled $\blacksquare$.

155. TACHS-A8-005 - Answer C: $\star$

$\star$ appears three times across the two groups, more often than $+$ or $\times$.

156. TACHS-A8-006 - Answer D: Four holes forming two horizontal mirrored pairs

Each punch reflects horizontally, giving two mirrored pairs and four holes.

157. TACHS-A8-007 - Answer C: Four holes in corresponding positions across both centerlines

Two perpendicular folds produce four positions reflected across both centerlines.

158. TACHS-A8-008 - Answer B: One symmetric diamond-shaped opening centered on the diagonal

The triangular cut on the fold reflects to form one symmetric diamond-like opening.

159. TACHS-A8-009 - Answer C: Six holes forming three vertical mirrored pairs

Each punch reflects vertically, producing three vertical pairs and six holes.

160. TACHS-A8-010 - Answer D: $\diamond\triangle\blacklozenge$

Each group has two different open shapes followed by the filled version of the first shape.

161. TACHS-A8-011 - Answer A: Left $\diamond$; center $\star$; right $\diamond$

Matching outer symbols surround a different open center shape.

162. TACHS-A8-012 - Answer A: Outer $\diamond$; inner $\circ$ at top

An open outer shape contains a different symbol positioned at the top.

163. TACHS-A8-013 - Answer B: $\blacklozenge|\blacksquare$

The prior right symbol moves left, while the filled version of the prior left symbol moves right.

164. TACHS-A8-014 - Answer D: $\square\leftarrow$

The arrow shifts behind the unchanged square and rotates clockwise from down to left.

165. TACHS-A8-015 - Answer C: $\triangle\diamond+$

Moving the middle plus sign to the end leaves the triangle and diamond in their original order.

166. TACHS-A8-016 - Answer A: Top $\blacklozenge$; Bottom $\triangle$

As in the top row, the two symbols exchange their upper and lower positions.

167. TACHS-A8-017 - Answer C: Two triangular cutouts mirrored across the fold diagonal

Unfolding reflects the cut across the single diagonal fold, creating two symmetric cutouts.

168. TACHS-A8-018 - Answer D: At the four outer corners of the original sheet

The final folded upper-left corner maps through the two centerline reflections to all four original outer corners.

169. TACHS-A8-019 - Answer C: Two holes on the top edge, equally spaced from the vertical centerline

The vertical fold reflects the punch to a second top-edge position at the same distance from the centerline.

170. TACHS-A8-020 - Answer D: Exactly four holes appear, related by the two fold reflections.

An interior punch off both folds doubles with each of the two reflections, producing four related locations.

171. TACHS-A8-021 - Answer B: $\star\blacklozenge\star$

Each group has matching open outer shapes surrounding a different filled center shape.

172. TACHS-A8-022 - Answer B: $\rightarrow\diamond\leftarrow$

The arrows on either side of the open center shape point in opposite directions.

173. TACHS-A8-023 - Answer C: $\diamond-\blacklozenge$

Like each example, this group places an open shape left and its filled counterpart right.

174. TACHS-A8-024 - Answer D: Top: $\diamond$; Center: $-$; Bottom: $\diamond$

Matching open shapes occupy the top and bottom positions around a different central symbol.

175. TACHS-A8-025 - Answer D: $\triangle$

Removing the repeated pair of diamonds leaves the single triangle.

176. TACHS-A8-026 - Answer C: Top-left $\uparrow$; Bottom-right $\square$

The arrow moves clockwise to the top-left and rotates from left to up, while the square moves counterclockwise to bottom-right.

177. TACHS-A8-027 - Answer A: $\blacktriangle \blacktriangle \blacktriangle \blacktriangle$

The top-right count of two doubles down the column, producing four filled triangles.

178. TACHS-A8-028 - Answer B: $\diamond$ inside $\triangle$

Exchanging the inner triangle and outer diamond places the diamond inside the triangle.

179. TACHS-A8-029 - Answer B: $+ |$

The repeated multiplication sign is removed, leaving the plus sign followed by the vertical line.

180. TACHS-A8-030 - Answer C: Eight holes related by three fold reflections

Each of three folds doubles an off-fold interior punch, producing $2 \times 2 \times 2 = 8$ holes.

181. TACHS-A8-031 - Answer B: Two matching notches on the curved edge, mirrored across the vertical diameter

Reversing the single fold reflects the notch to the opposite side of the circle's curved edge.

182. TACHS-A8-032 - Answer A: Two holes on the vertical centerline, mirrored across the horizontal centerline

The horizontal fold doubles the hole, while its position on the vertical centerline remains unchanged.

183. TACHS-A8-033 - Answer D: 4

Each of the two folds doubles an interior off-fold punch, producing $2 \times 2 = 4$ holes.

184. TACHS-A8-034 - Answer A: $\blacklozenge \leftarrow \star$

The left-pointing arrow correctly points toward the filled diamond on its left.

185. TACHS-A8-035 - Answer A: $\leftarrow \uparrow \rightarrow$

Each arrow is obtained by rotating the preceding arrow 90° clockwise.

186. TACHS-A8-036 - Answer C: Top-left: $\square$; Top-right: $\blacktriangle$; Bottom: $\triangle$

The top-right shape is the filled counterpart of the different open shape at the bottom.

187. TACHS-A8-037 - Answer B: Top: $\square \diamond$; Bottom: $\blacklozenge \blacksquare$

The bottom row contains the filled counterparts of the top row in reverse order.

188. TACHS-A8-038 - Answer D: $+ \times |$

Combining the two cells and writing the shared multiplication sign once produces plus, multiplication, vertical line.

189. TACHS-A8-039 - Answer A: $\diamond | \star$

Advancing the triangle and diamond one place in the stated cycle produces diamond and star.

190. TACHS-A8-040 - Answer C: $\blacksquare \blacksquare$

Removing one of three open squares leaves two shapes, and changing fill produces two filled squares.

191. TACHS-A8-041 - Answer A: $\triangle + \diamond$

Rotating the center multiplication sign 45° clockwise produces a plus sign while the outer shapes remain fixed.

192. TACHS-A8-042 - Answer C: Three holes, one in each vertical panel at matching heights

The three stacked panels separate during unfolding, placing one corresponding hole in each panel.

193. TACHS-A8-043 - Answer B: Two matching notches on the left edge, mirrored across the horizontal centerline

Reversing the horizontal fold reflects the edge notch to a second position on the same outer edge.

194. TACHS-A8-044 - Answer A: One hole located on the diagonal fold line

A punch exactly on a fold maps onto itself when the sheet is reopened.

195. TACHS-A8-045 - Answer B: 4

Each off-fold punch reflects once across the vertical centerline, so two punches produce four holes.

196. TACHS-A8-046 - Answer D: Top-left: $-$; Top-right: $|$; Bottom: $+$

A horizontal and a vertical line combine to form the plus sign below them.

197. TACHS-A8-047 - Answer B: $\square \blacksquare \triangle \blacktriangle$

Each open shape is immediately followed by its filled counterpart, and the pattern repeats with a different shape.

198. TACHS-A8-048 - Answer A: Top-left: $\square$; Top-right: $\blacktriangle$; Bottom-left: $\triangle$; Bottom-right: $\blacksquare$

Open shapes occupy the left column, and their filled counterparts appear diagonally opposite in the right column.

199. TACHS-A8-049 - Answer A: $\leftarrow \leftarrow \rightarrow$

Two identical arrows are followed by one arrow pointing in the opposite direction.

200. TACHS-A8-050 - Answer B: + ▲ △

The center filled triangle matches the open triangle on the right, while the left plus sign is different.